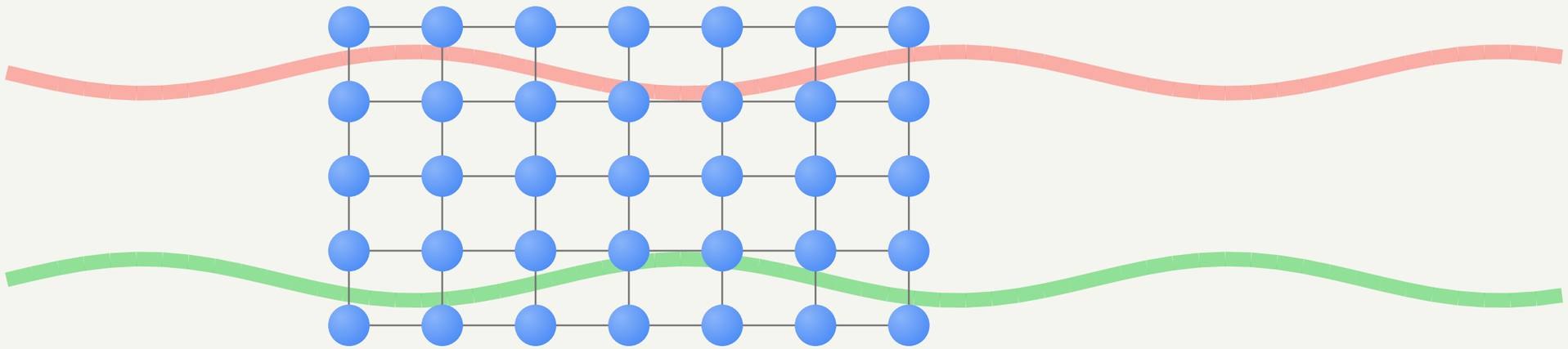
Overcoming the Speed Limit of Quantum Error Correction in Neutral Atom Arrays

A Measurement-Free and Movement-Free Protocol Using Multi-Species Atoms and Global Control



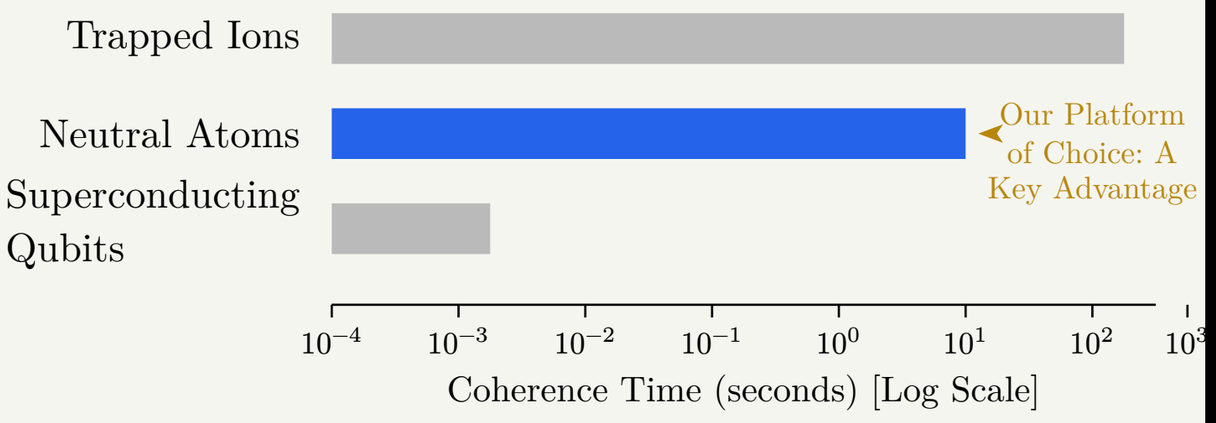
This work presents a novel QEC scheme that eliminates the dominant latency sources in
neutral atom architectures, accelerating the path to fault-tolerant quantum computation.

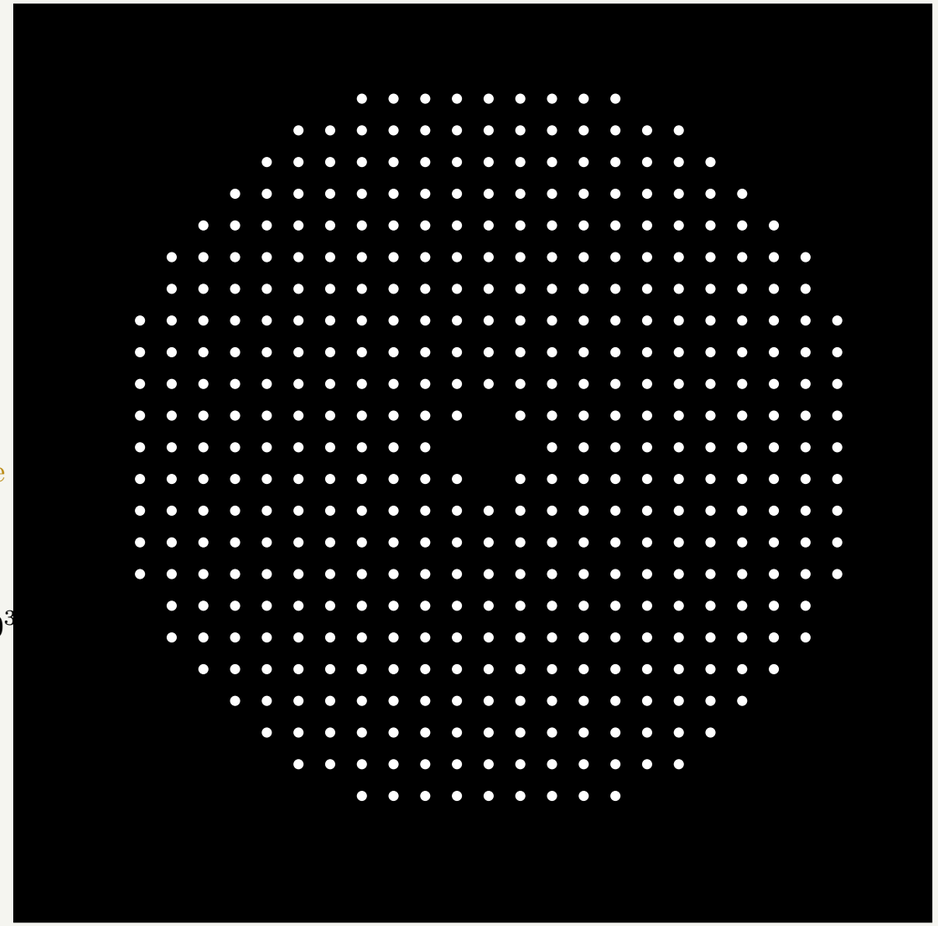
Neutral atom arrays have emerged as a leading platform for scalable quantum computing.

Key strengths that make this platform exceptionally promising:

- **Exceptional Scalability:** Demonstrations of arrays with thousands of atomic qubits.
- **Long Coherence Times:** Qubit coherence can surpass one second, significantly longer than many competing platforms.
- **High Gate Fidelity:** Two-qubit gate fidelities have been pushed beyond 99.5%.

- **Strong, Tunable Interactions:** Rydberg states enable strong, controllable interactions between atoms.





QEC cycles are orders of magnitude slower than quantum gates, creating a critical performance bottleneck.

Gate Operation
(1 μs)

Measurement &
Movement Process

Extended Duration
(hundreds of μs)

Quantum Gates:

Corrective Actions: Conventional QEC requires operations that are dramatically slower:

Operations are incredibly fast, taking approximately 1 microsecond (μs).

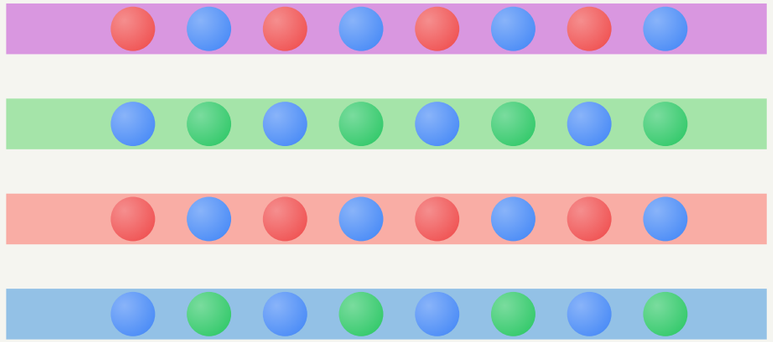
- Mid-circuit Measurement: 100–500 μs (fluorescence detection)
- Atom Shuttling/Movement: 10s–100s of μs

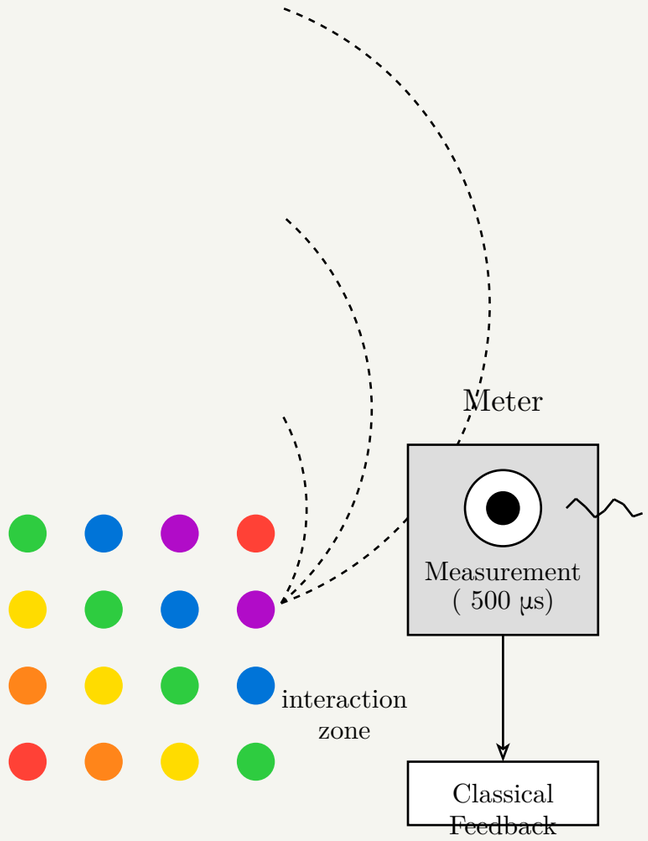
The Consequence: The quantum state decoheres while waiting for slow measurements and movements, undermining the very purpose of error correction.

Our solution is a new QEC paradigm that is both measurement-free and movement-free.

Conventional QEC

Our Protocol





Measurement-Free: Replaces slow, destructive fluorescence measurements with fully coherent operations. Syndrome information is mapped to ancilla qubits, which then trigger

Movement-Free: Eliminates the need for atom shuttling. The atoms remain in a static lattice, and all interactions are mediated by precisely shaped, global laser pulses.

corrective gates without classical feedback.

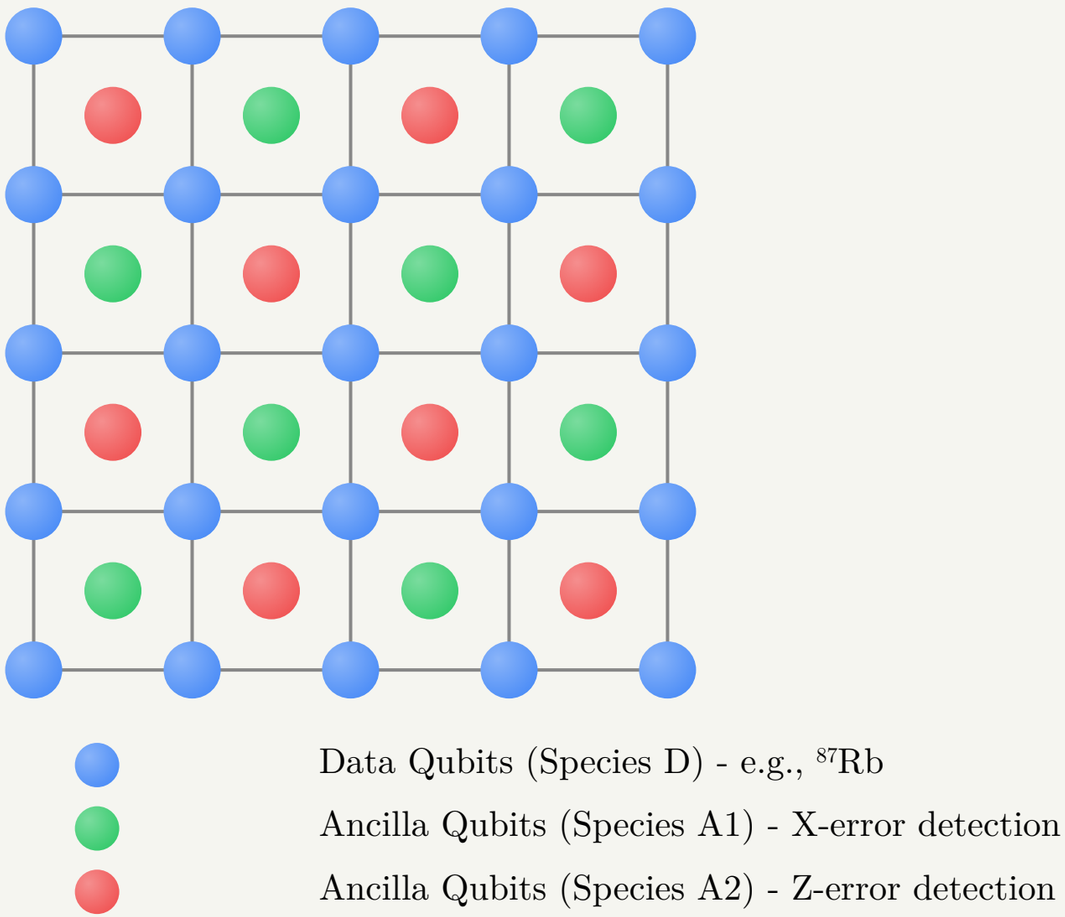
Key Insight: By executing the entire QEC cycle—syndrome mapping, error correction, and ancilla reset—unitarily, we remove the primary sources of latency and decoherence.

The protocol is implemented on a static, three-species Rydberg atom array.

The lattice architecture is composed of three distinct roles:

- **Data Qubits (Species D):** Encode the logical information, located at the vertices of a square lattice.
- **Ancilla Qubits (Species A1):** Used to detect X-type errors, located at the centers of the 'X-stabilizer' plaquettes.
- **Ancilla Qubits (Species A2):** Used to detect Z-type errors, located at the centers of the 'Z-stabilizer' plaquettes.

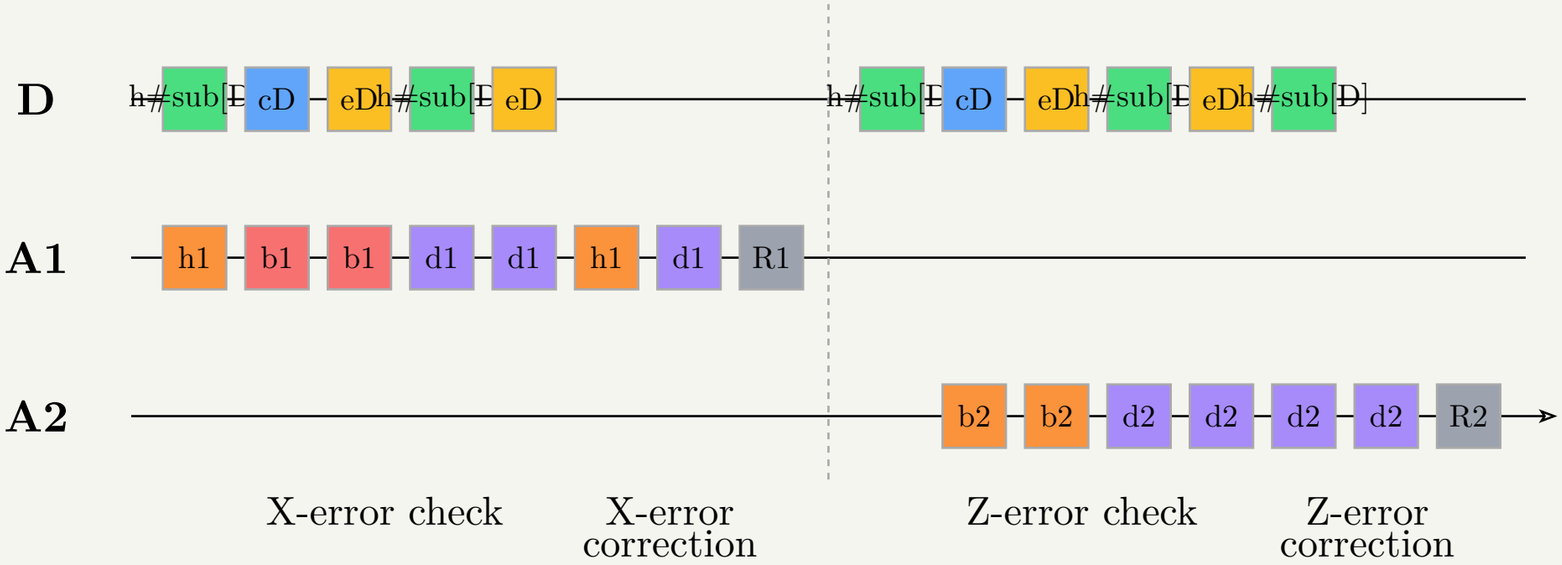
- **Species-Selective Control:** The distinct atomic species have well-separated transition frequencies, allowing global laser pulses to address one species without affecting the others.



A complete QEC cycle is driven by a sequence of global, species-selective laser pulses.

The full cycle for one error type is executed in two main phases:

- **Error Check:** Parity information from four data qubits is coherently mapped onto a central A1 ancilla. This involves a sequence of Hadamard gates and



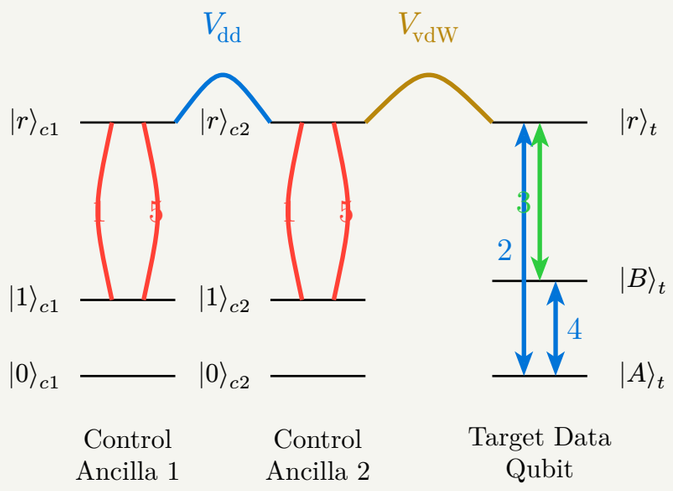
conditional multi-qubit gates (OR gate, C_2NOT).

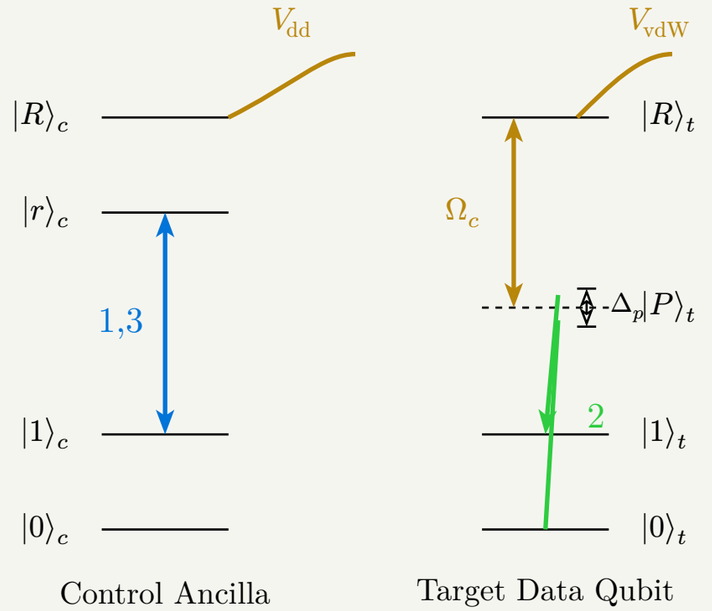
- **Error Correction:** The resulting state of the A1 ancillas then controls a final C_2NOT gate, which conditionally flips the erroneous data qubit to correct the error.
- **Reset:** The A1 ancillas are reset to $|0\rangle$ to prepare for the next cycle. An analogous sequence using A2 ancillas corrects Z-errors.

Multi-qubit gates are realized using Electromagnetically Induced Transparency and Rydberg blockade.

OR Gate (via EIT)

CCNOT (C_2NOT) Gate





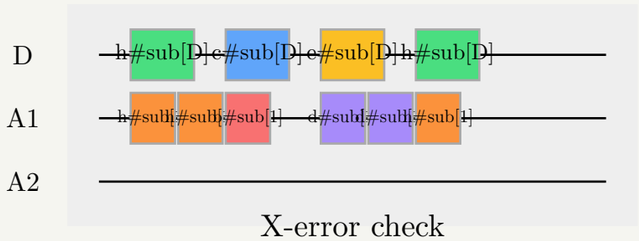
The data qubit is flipped only if *both* neighboring ancillas are in a specific state. This is achieved by configuring the Rydberg blockade such that the presence of ancillas in a blocking state prevents the transition on the data qubit.

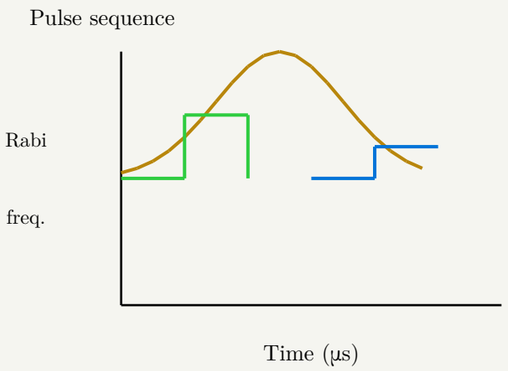
An ancilla in the $|1\rangle$ state is excited to a Rydberg state. This shifts the energy levels of the neighboring data qubit, breaking an Electromagnetically Induced Transparency (EIT) condition and allowing a Raman transition (a flip) to occur.

Mechanism: The protocol relies on strong Rydberg interactions between data and ancilla qubits (V_{dd}) being much stronger than between data qubits themselves (V_{vdW}).

Pulse-level simulations on a 9-atom unit cell validate the high fidelity of the parity-check mechanism.

To verify the physical implementation, we performed exact numerical integration of the Schrödinger equation for a 9-atom cluster (4 data, 5 ancilla qubits).



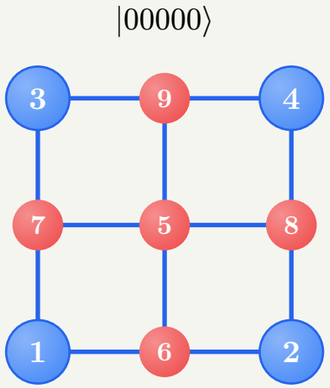


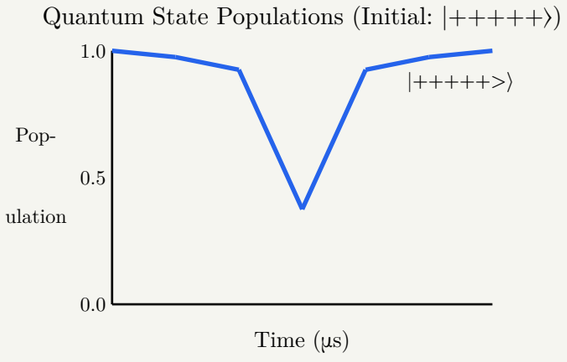
Scenario 1: No Error

When the system starts in an error-free state (e.g., $|+++++\rangle$), the parity-check sequence correctly leaves the ancillas in

the $|0\rangle$ state and returns the data qubits to their original state.

Fidelity: Simulations confirm individual operation fidelities exceeding 99%, limited primarily by non-adiabatic effects.



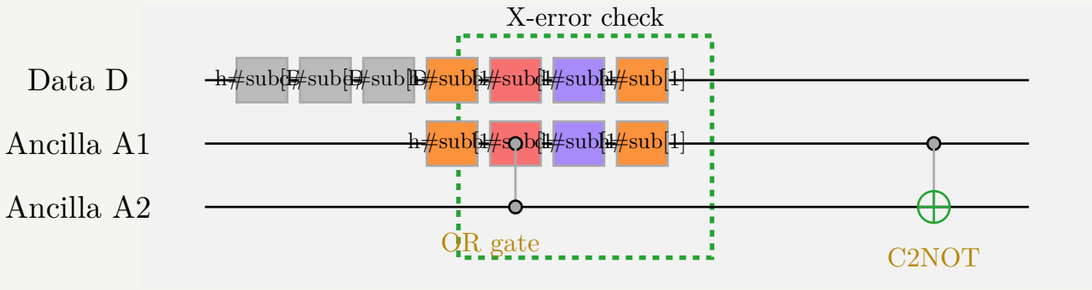


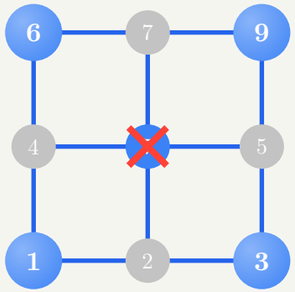
The protocol correctly identifies a single-qubit error and triggers the appropriate correction.

Scenario 2: Single Error

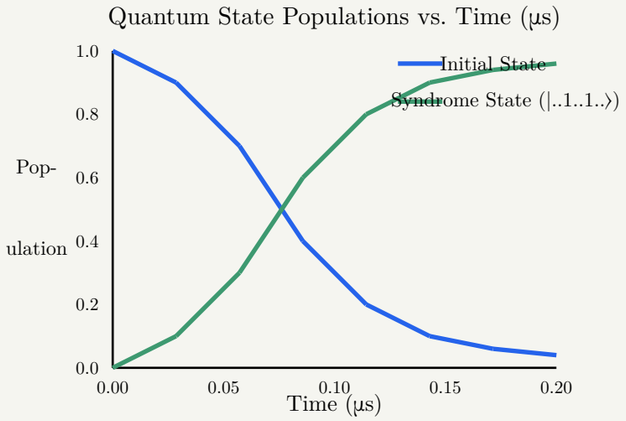
We introduce a single X-error (a bit flip) on one data qubit.

Detection: After the parity-check sequence, the two ancillas neighboring the flipped qubit correctly transition to the $|1\rangle$ state, flagging the precise location of the error. All other ancillas remain in $|0\rangle$.





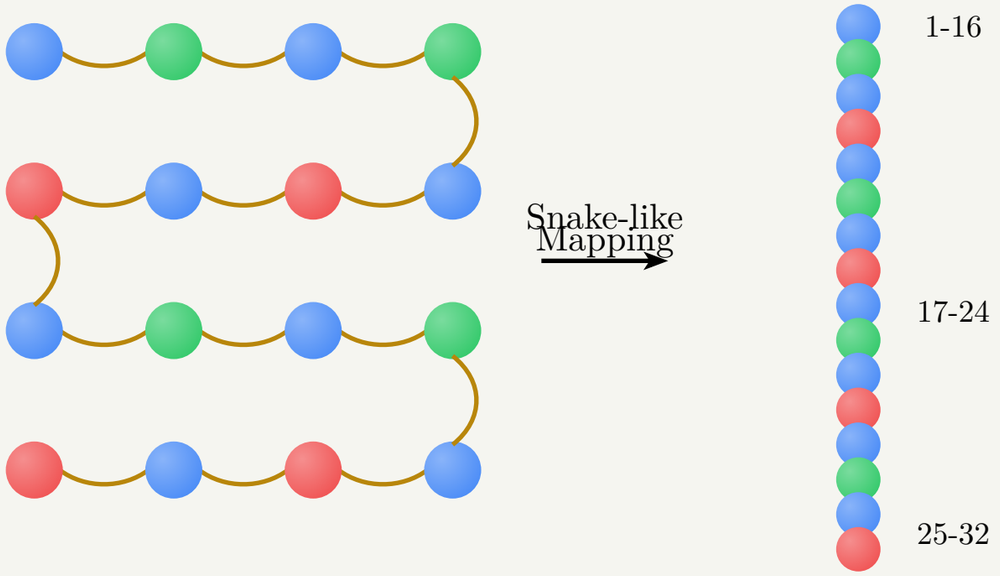
Correction: The state of these two ancillas then triggers the double-controlled correction gate, which successfully flips the data qubit back to its correct state.



Tensor network simulations are used to verify the protocol's performance on scalable quantum systems.

The Challenge: Simulating large quantum systems is classically intractable due to the exponential growth of the Hilbert space.

The Method: We use Matrix Product States (MPS), a type of tensor network, to efficiently represent the quantum state. This method accurately captures



the system's entanglement structure, allowing for simulations of larger arrays.

Implementation: The 2D atom array is mapped to a 1D chain for the MPS simulation. We use the time-dependent variational principle (TDVP) to accurately evolve the state under the QEC pulse sequence.

Active error correction maintains >99.998% ground state probability, while uncorrected states rapidly decay.

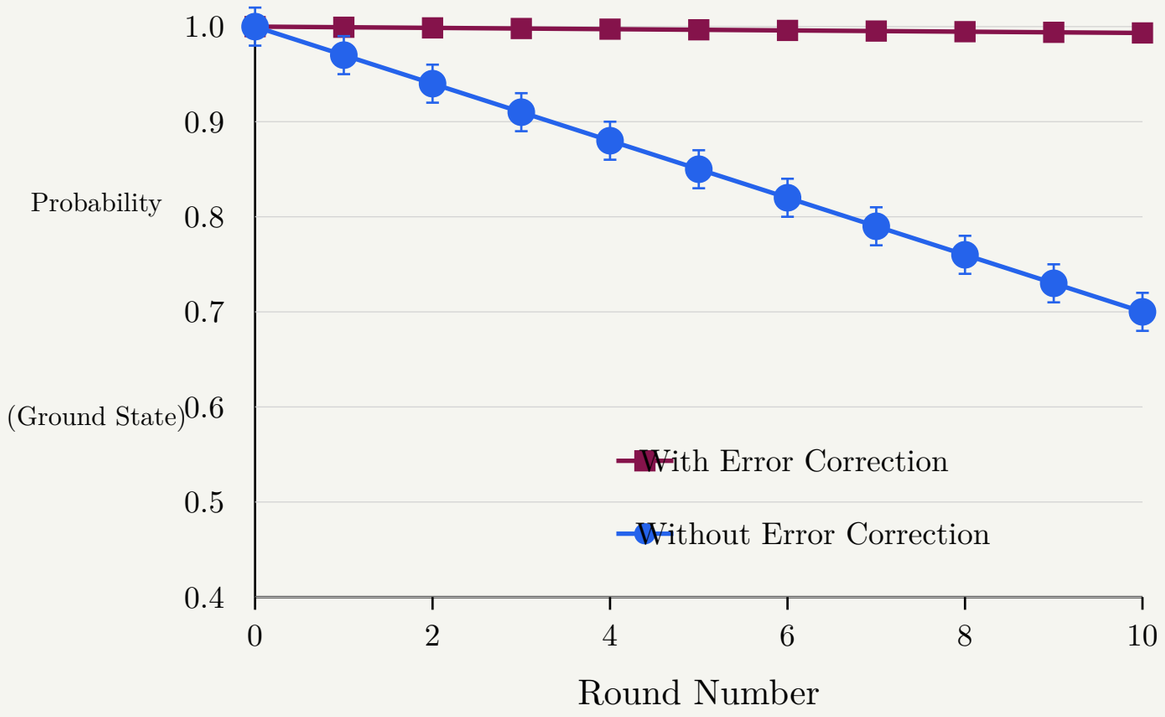
We simulated 10 full QEC cycles on a 4x4 toric code (32 qubits), applying random Pauli errors each cycle. The results show a stark contrast:

- **With Correction (Red Line):** The system's probability of being in the correct ground state remains near unity. The protocol effectively finds

Ground State Probability: With vs. Without Error Correction

and fixes errors, preserving the logical information.

- **Without Correction (Blue Line):** Errors accumulate unchecked. The ground state probability decays steadily to approximately 60% after just 10 cycles, indicating significant information loss.



The measurement-free approach offers a 1-2 order of magnitude speedup in QEC cycle time.

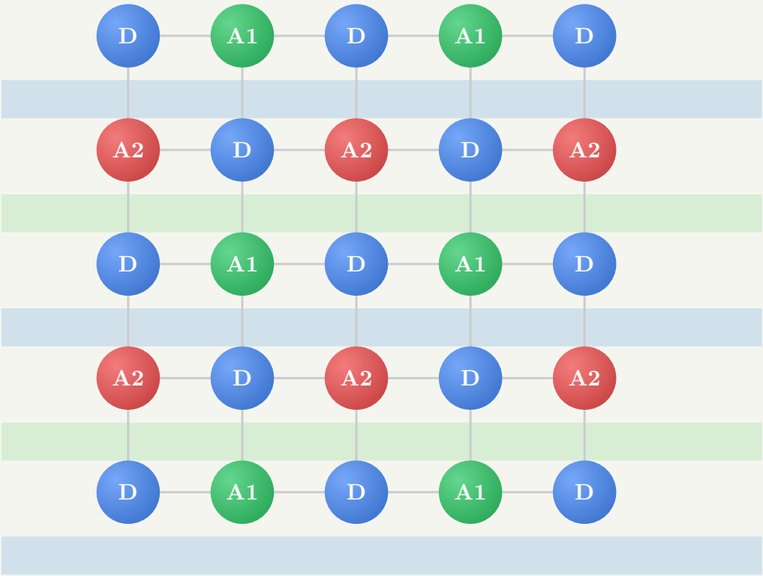
Feature	Conventional Measurement-Based QEC	This Work: Measurement-Free QEC
Syndrome Readout	Slow (100–500 μs)	Coherent & Fast (< 10 μs)
Atom Shuttling	Required	**Not required** (Static Array)
Local Addressing	Required (Complex hardware)	**Not required** (Global Pulses)
Classical Feedback	Required	**Not required** (Autonomous)

Cycle Time	1 ms	**10–100 μs**
Ancilla Overhead	1 auxiliary species	2 auxiliary species

The tradeoff of requiring a second ancilla species is vastly outweighed by the dramatic reduction in cycle time and hardware complexity.

This protocol provides a viable pathway toward hardware-efficient, fault-tolerant quantum computation.

- We have designed and validated a measurement-free and movement-free QEC protocol for the toric code.
- By combining a multi-species architecture with global, species-selective pulses, the scheme eliminates the primary



- Pulse-level and tensor network simulations confirm the protocol's ability to coherently detect and autonomously correct errors, maintaining high fidelity over many cycles.
- This approach significantly reduces QEC

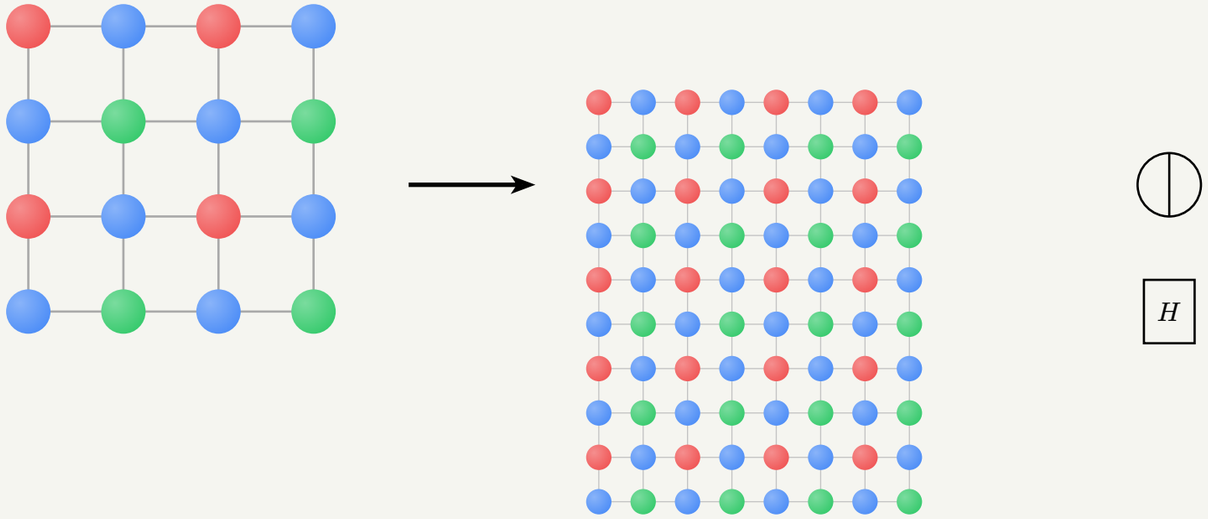
bottlenecks of measurement latency and atom shuttling.

cycle times and simplifies experimental complexity, accelerating progress in neutral atom quantum computing.

The measurement-free paradigm opens several compelling directions for future research.

Key areas for future work include:

- **Scaling to Larger Codes:** Demonstrating error suppression with higher-distance toric codes.
- **Exploring Advanced Codes:** Adapting the protocol to other promising code families like concatenated or LDPC codes.



- **Rigorous Fault-Tolerance Analysis:** Performing a full noise threshold analysis under realistic experimental conditions.
- **Implementing Logical Gates:** Extending the framework to perform universal logical operations on the encoded qubits.
- **Near-Term Experimental Demonstration:** Realizing the protocol in existing dual-species Rydberg atom arrays.